Classification Models, Evaluation Metrics & Handling Imbalanced Data

Main Crafts AI & Machine Learning Internship Task 4

BY: Choudavaram Sree Harshith Reddy

B.Tech – CSE (AI & ML)

1. Introduction

The objective of this project is to build and evaluate a binary classification model using machine learning techniques. This project focuses on understanding classification problems and evaluating models using appropriate metrics beyond accuracy.

Classification plays a crucial role in real-world applications such as fraud detection, spam filtering, medical diagnosis, and customer churn prediction. In this project, Logistic Regression is used as the primary model, and its performance is evaluated using metrics such as precision, recall, F1-score, and ROC-AUC.

Additionally, the project explores handling class imbalance and compares the performance of multiple models.

2. Dataset Description

The dataset used in this project is the Breast Cancer dataset provided by the Scikit-learn library. It is widely used for binary classification tasks.

Dataset Details:

- Total samples: 569
- Features: 30 numerical features
- Target classes:
- 0 → Malignant (cancerous)
- 1 → Benign (non-cancerous)

The dataset is slightly imbalanced, which makes it suitable for studying classification metrics and imbalance handling techniques.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset structure, feature distributions, and class balance.

EDA Steps Performed:

- Checked dataset shape and basic statistics
- Analyzed class distribution
- Observed feature variations

Observations:

- The dataset contains slightly more benign cases than malignant cases
- Features are on different scales, requiring normalization
- No missing values are present

4. Data Preprocessing

Before training the model, the dataset was prepared using the following steps:

- Splitting dataset into training (80%) and testing (20%) sets
- Using stratification to maintain class balance
- Applying StandardScaler for feature normalization

Feature scaling is important for models like Logistic Regression to ensure better convergence and performance.

5. Model Training

A Logistic Regression model was used as the baseline classifier.

- The model was trained using scaled training data
- Maximum iterations were set to 1000 to ensure convergence

This model predicts whether a tumor is malignant or benign based on input features.

6. Model Evaluation

The model was evaluated using confusion matrix and classification metrics.

Confusion Matrix Explanation:

- True Positive (TP): Correctly predicted malignant cases
- True Negative (TN): Correctly predicted benign cases
- False Positive (FP): Incorrectly predicted malignant
- False Negative (FN): Missed malignant cases

Classification Metrics:

- Precision: Measures correctness of positive predictions
- Recall: Measures ability to detect actual positives
- F1-Score: Balance between precision and recall

Observation:

- The model performs well with high precision and recall
- F1-score indicates balanced performance

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Confusion Matrix:
 [[41  1]
 [ 1 71]]

Classification Report:

              precision    recall  f1-score   support

           0       0.98      0.98      0.98        42
           1       0.99      0.99      0.99        72

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114
```

7. ROC Curve & AUC

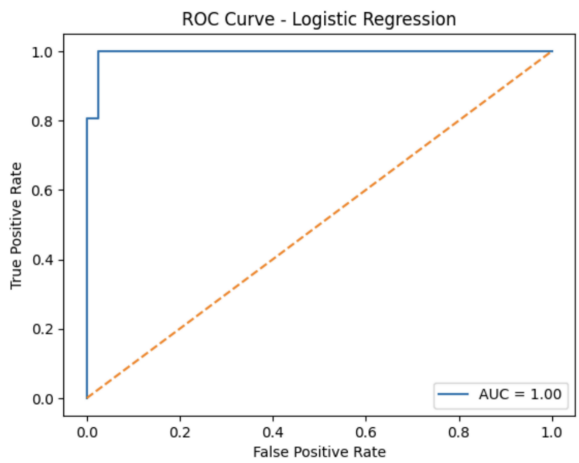
The ROC curve was plotted to evaluate model performance across different thresholds.

- X-axis → False Positive Rate
- Y-axis → True Positive Rate

AUC (Area Under Curve) measures how well the model distinguishes between classes.

Observation:

- AUC value is close to 1, indicating strong classification ability



8. Handling Class Imbalance

Class imbalance occurs when one class has more samples than the other.

To address this:

- Logistic Regression with class_weight="balanced" was used

Observation:

- Improved recall for minority class
- Better overall F1-score

9. Model Comparison

Two models were compared:

Logistic Regression

- Stable and consistent
- Less prone to overfitting

Decision Tree

- Easy to interpret
- Can overfit training data

Observation:

- Logistic Regression showed better generalization
- Decision Tree showed higher variance

10. Results & Visualization

The performance of models was compared using classification reports.

Observations:

- Logistic Regression achieved balanced performance
- Decision Tree may overfit without tuning
- Class balancing improved minority class detection

11. Conclusion & Future Scope

In this project, a binary classification model was successfully built and evaluated using multiple performance metrics.

Key learnings:

- Accuracy alone is not sufficient for evaluation
- Precision, Recall, and F1-score are essential
- ROC-AUC provides better performance insight
- Handling imbalance improves model quality

Future Improvements:

- Hyperparameter tuning
- Use advanced models (Random Forest, XGBoost)
- Use larger datasets
- Deploy model as a web application

Decision Tree Report:

	precision	recall	f1-score	support
0	0.85	0.93	0.89	42
1	0.96	0.90	0.93	72
accuracy			0.91	114
macro avg	0.90	0.92	0.91	114
weighted avg	0.92	0.91	0.91	114

Balanced Model Report:

	precision	recall	f1-score	support
0	0.91	0.98	0.94	42
1	0.99	0.94	0.96	72
accuracy			0.96	114
macro avg	0.95	0.96	0.95	114
weighted avg	0.96	0.96	0.96	114